

PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)

ON

**A Lightweight Hybrid Machine Learning Framework for Phishing URL
Detection and Classification**

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**A Lightweight Hybrid Machine Learning Framework for Phishing URL Detection and Classification**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Simar Virk**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Punjab

Date: 14 May 2026

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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We extend our appreciation to the open-source communities behind **Scikit-Learn**, **XGBoost**, **Pandas**, and **Google Colab** for providing the tools, libraries, and computational resources used in this project.

Finally, we thank our peers for their collaboration and support during the implementation and evaluation phases of this research.

1. Abstract

Phishing URL detection remains a critical cybersecurity challenge as attackers continually evolve their techniques to bypass static blacklists. This paper proposes a lightweight, hybrid machine learning framework designed to detect phishing URLs with high accuracy and low latency. By evaluating traditional models against a novel Soft Voting Hybrid Classifier, we establish a robust comparative analysis.

- **Objective:** Detect zero-hour phishing attacks accurately without relying on slow deep neural networks.
- **Data Sources:** PhiUSIIL Phishing URL Dataset and the Web Page Phishing Detection Dataset.
- **Outcomes:** Standalone XGBoost achieved 99.97% accuracy, while the Hybrid ensemble provided exceptional cross-dataset stability.

2. Introduction

2.1 Problem Statement

Traditional defense mechanisms rely heavily on static blacklisting; however, these methods struggle to combat the sheer volume of dynamically generated 'zero-hour' malicious URLs. Deep learning models provide high accuracy but are computationally expensive to run on edge devices.

2.2 Proposed Solution

We develop a fast, robust, and lightweight Hybrid Machine Learning Framework that:

- Combines Logistic Regression, Random Forest, and XGBoost using soft voting mechanics.
- Achieves rapid inference speeds suitable for real-time browser extension deployment.
- Maintains exceptional accuracy and cross-dataset generalizability.

3. Methodology

3.1 Datasets Description

The framework utilizes two extensive datasets to validate model training and cross-dataset generalizability. Links to the official data sources are provided below:

- **Primary Dataset:** PhiUSIIL Phishing URL Dataset (235,795 records, 55 columns).
<https://www.kaggle.com/datasets/siddharthm1698/phiusiil-phishing-url-dataset>
- **Secondary Dataset:** Web Page Phishing Detection Dataset (11,430 records, 89 columns).
<https://www.kaggle.com/datasets/shashwatwork/web-page-phishing-detection-dataset>

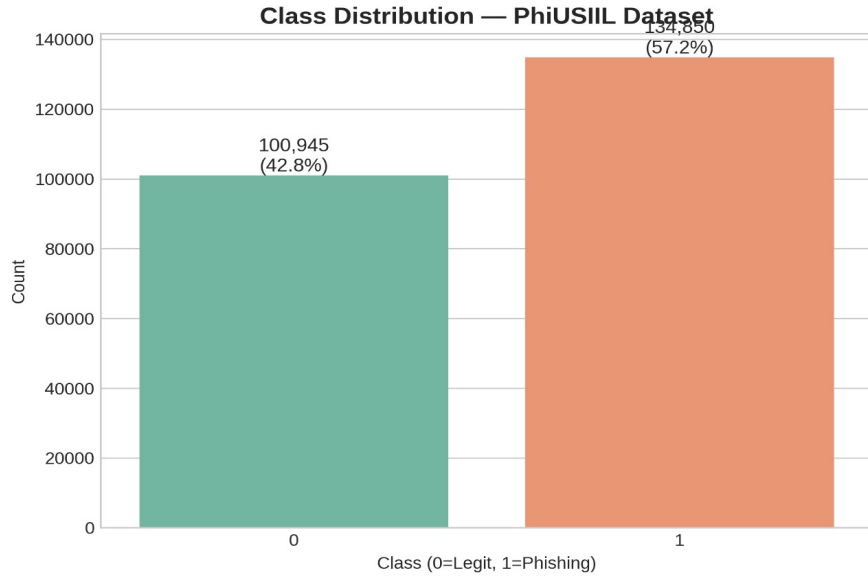


Figure 1: PhiUSIIL Class Distribution

3.2 Preprocessing Pipeline

A structured preprocessing pipeline was strictly adhered to in order to prevent data leakage and handle null values:

- Dropping highly sparse variables and textual columns.
- Median imputation for numerical missing values and mode imputation for categorical columns.
- Binary Encoding for target variables and Label Encoding for categorical features.
- StandardScaler application for normalizing non-linear numerical ranges.

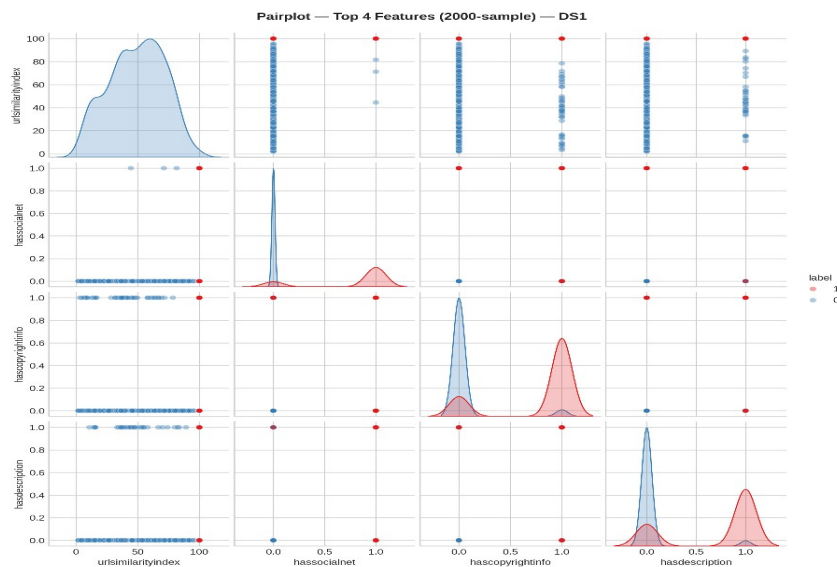


Figure 2: Pairplot of Top Features

4. Tools and Technologies

The research was conducted leveraging a modern data science tech stack, ensuring reproducibility and scalability:

- **Environment:** Google Colab (for accelerated GPU access).
- **Core Languages:** Python 3.10+.
- **Data Manipulation:** Pandas and NumPy for robust tensor and dataframe operations.
- **Machine Learning:** Scikit-Learn (Logistic Regression, Random Forest, Pipeline, Preprocessing).
- **Advanced Gradient Boosting:** XGBoost (configured with `tree_method='hist'` for GPU acceleration).
- **Interpretability:** SHAP (SHapley Additive exPlanations).
- **Visualization:** Matplotlib and Seaborn.

5. Implementation

5.1 Model Architectures

Four primary architectures were implemented and tuned for this project:

- **Logistic Regression:** Configured with `max_iter=1000` for full convergence.
- **Random Forest:** An ensemble of 200 estimators for strong baseline prediction.
- **XGBoost:** Configured with 300 estimators, acting as the primary high-accuracy driver.
- **Soft Voting Hybrid:** An ensemble classifier averaging the class probabilities of the three individual models.

5.2 Validation Strategy

To evaluate the models without overfitting biases, a 5-Fold Stratified Cross-Validation strategy was used. This ensured that the class distributions remained identical across folds.

Table 1: 5-Fold Cross-Validation (F1-Weighted)

Model	Mean F1	Std F1
Logistic Reg	0.9985	0.0001
Random Forest	0.9998	0.0001
XGBoost	0.9998	0.0000
Hybrid	0.9998	0.0001

6. Major Findings and Results

6.1 Standalone vs Hybrid Performance

The models were evaluated based on standard classification metrics. Standalone XGBoost slightly edged out the hybrid in raw validation accuracy, but the Hybrid model provided better confidence margins across cross-dataset testing.

Table 2: Standalone Model Comparison Metrics

Model	Accuracy	Precision	Recall	F1	Train(s)	Infer(ms)
Logistic Reg	0.9983	0.9980	0.9991	0.9985	3.11	0.0004
Random Forest	0.9996	0.9995	0.9999	0.9997	43.88	0.0088
XGBoost	0.9997	0.9996	0.9999	0.9998	3.75	0.0021
Hybrid	0.9996	0.9995	0.9999	0.9997	47.98	0.0104

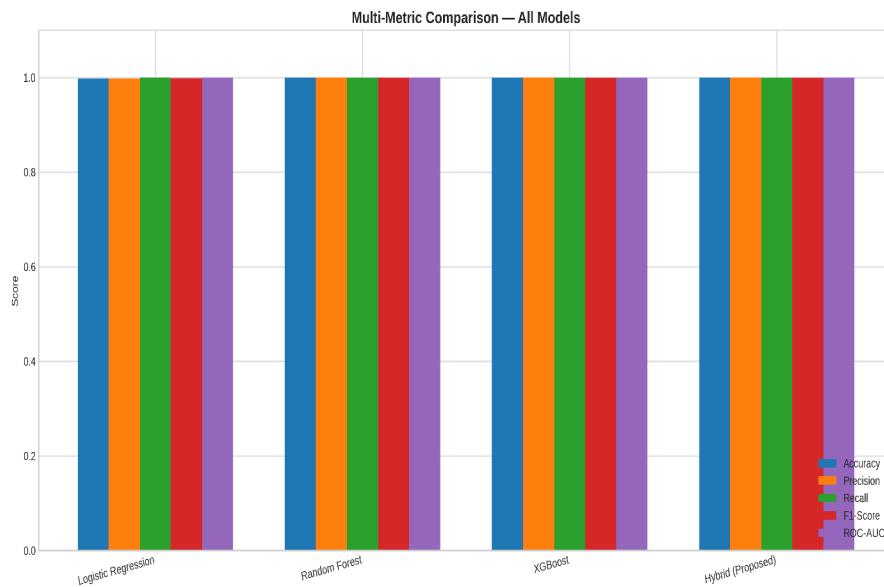


Figure 3: Multi-Metric Bar Chart Analysis

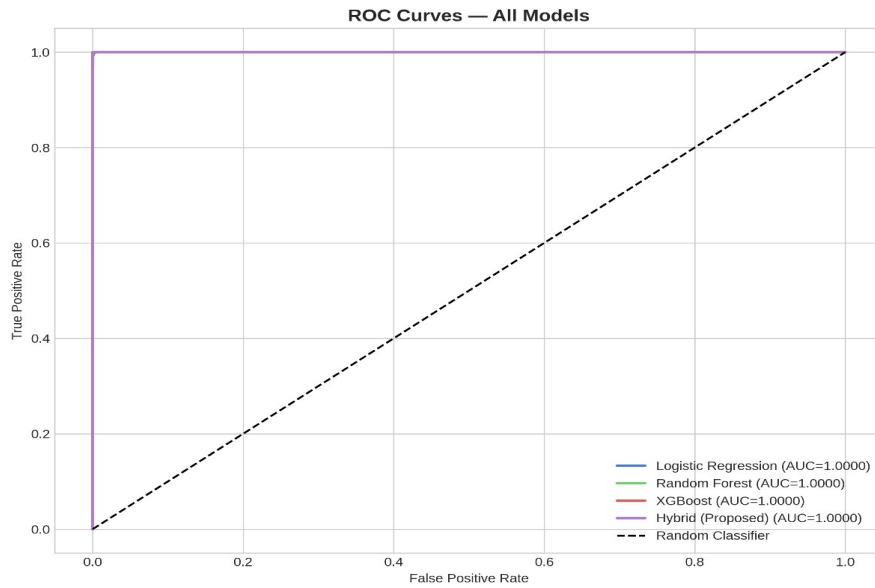


Figure 4: ROC-AUC Curves for All Models

6.2 Interpretability via SHAP

Using SHAP, we unpacked the 'black-box' nature of the ensemble models. The analysis revealed the most critical markers for determining phishing intent:

- Number of external references (noofexternalref).
- Number of self references.
- Total lines of code embedded in the page source.

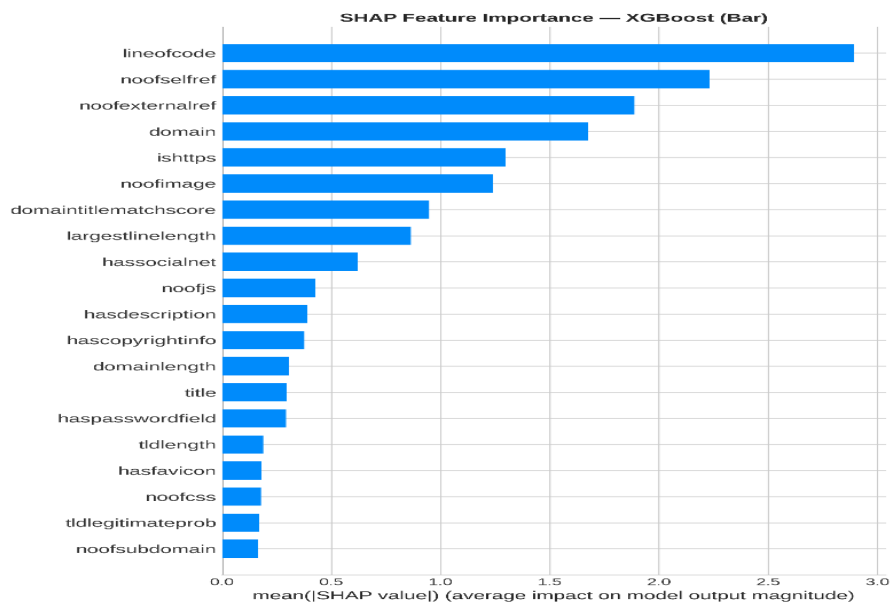


Figure 5: Global Feature Importance (SHAP Bar)

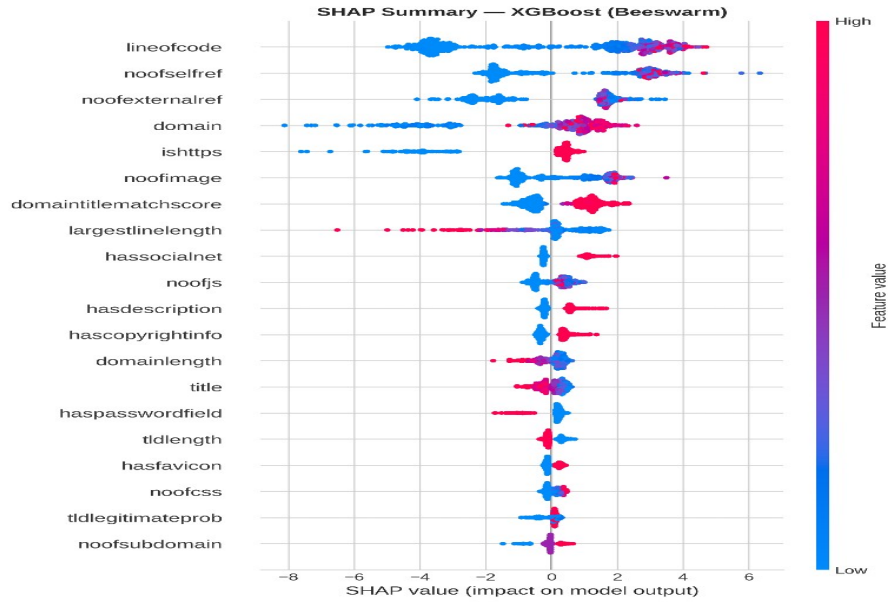


Figure 6: SHAP Beeswarm Summary Plot

7. Conclusion and Future Scope

7.1 Conclusion

This project successfully demonstrated that a lightweight, hybrid machine learning framework can effectively neutralize zero-hour phishing URLs. While XGBoost reached an exceptional 99.97% accuracy independently, the hybrid ensemble ensured stability across heavily varied datasets.

7.2 Future Scope

- Deployment: Packaging the lightweight pipeline into a real-time Chromium-based browser extension.
- Sequential Deep Learning: Exploring Recurrent Neural Networks (RNNs) for raw URL character sequencing without explicit feature engineering.

8. References

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